Software Testing & Quality Assurance Plan

Project: ERP Item Management System

Prepared by: Sahil Kumar (Github: https://github.com/Sahilll10)

1. Overview

The goal of this testing phase was to ensure the ERP system is reliable, secure, and handles real-world data entry errors gracefully. I performed a mix of **Functional Testing** (does it work?) and **Negative Testing** (does it break if the user makes a mistake?).

2. Functional Test Cases

These tests confirm that the core requirements of the assessment are fully met.

ID	Feature	Test Step	Expected Result
TC-01	**Secure Login**	Enter admin@admsystems.net and password123.	User is granted access to the Dashboard.
TC-02	**Item Entry**	Create a new item: "Aluminum Coil" with weight 250.75.	Item is saved and appears in the "Raw" status list.
TC-03	**Search Filter**	Search for "Alum" in the search bar.	The list filters to show only the Aluminum entry.
TC-04	**Item Processing**	Select "Process" and split 1 item into 2 child components.	The parent status updates to "Processed" and children are created.
TC-05	**Tree Traceability**	Open the "View Hierarchy Tree" page.	A visual map correctly links the parent to its children.

3. Edge Case & Boundary Testing

I specifically tested "stress points" where users might accidentally input incorrect data.

ID	Scenario	Test Step	Expected Result
TC-06	**Empty Fields**	Try to save an item with a blank name or weight.	Front-end validation stops the save and prompts the user.
TC-07	**Zero/Negative Weight**	Attempt to enter a weight of 0 or -10.5.	System rejects the entry; weights must be positive values.
TC-08	**Session Security**	Try to go to /Item/Index without logging in.	System blocks access and redirects to the Login page.
TC-09	**Search "No Match"**	Search for a random string like "123456789".	Table displays a "No items found" message instead of crashing.
TC-10	**Accidental Deletion**	Click the "Delete" button on an inventory item.	A browser confirmation pops up to prevent accidental data loss.

4. Technical Environment & Compatibility

- **Browser Testing:** Verified the UI layout and the "Tree View" lines in both Google Chrome and Microsoft Edge.
- **Responsive Design:** Used Bootstrap’s grid system to ensure the navigation bar and data tables remain readable on mobile devices and tablets.